PARIS HOUSE PRICE

Course: MIS 311

Quarter: 2 / 2025 - 2026

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Background

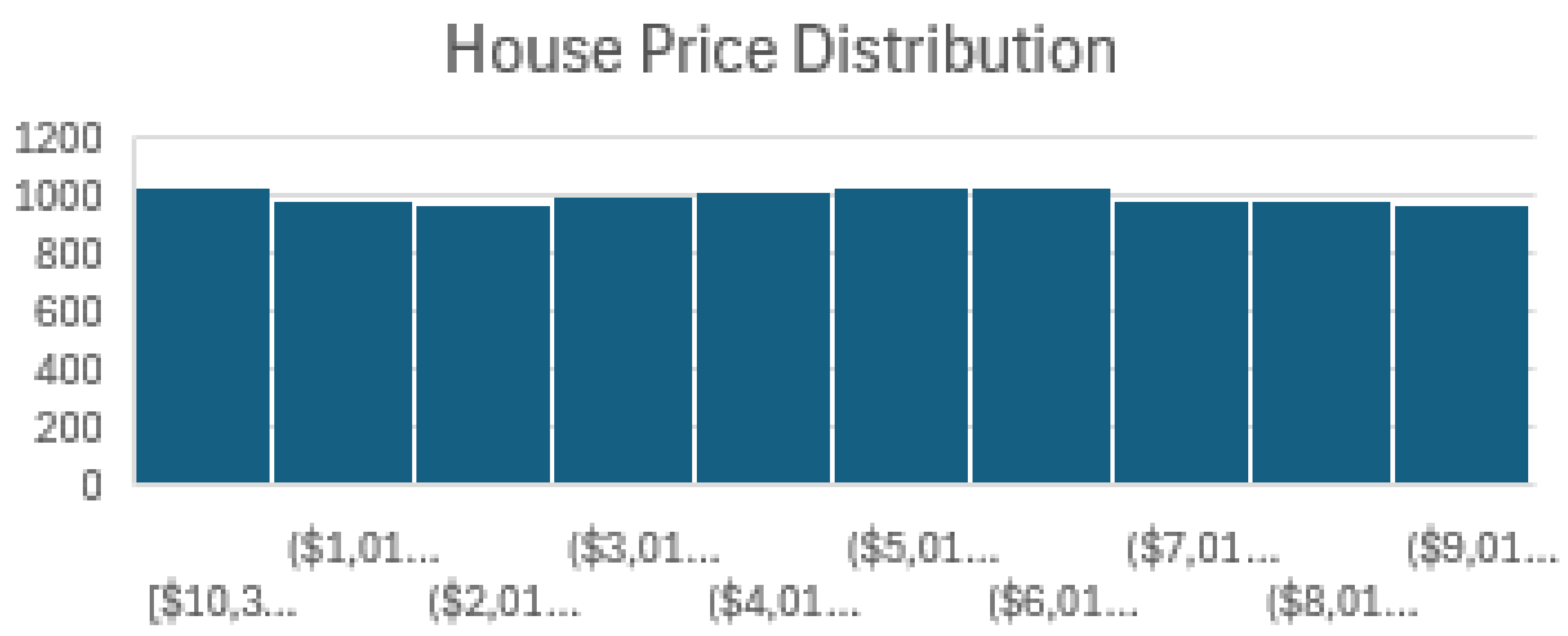
The Paris Market: One of the world's most premium and volatile real estate hubs.
Complexity: Property values are driven by a diverse set of traditional and luxury features.
Significance: Accurate valuation is critical for investors, urban planners, and homebuyers to mitigate financial risks.

Dataset Overview

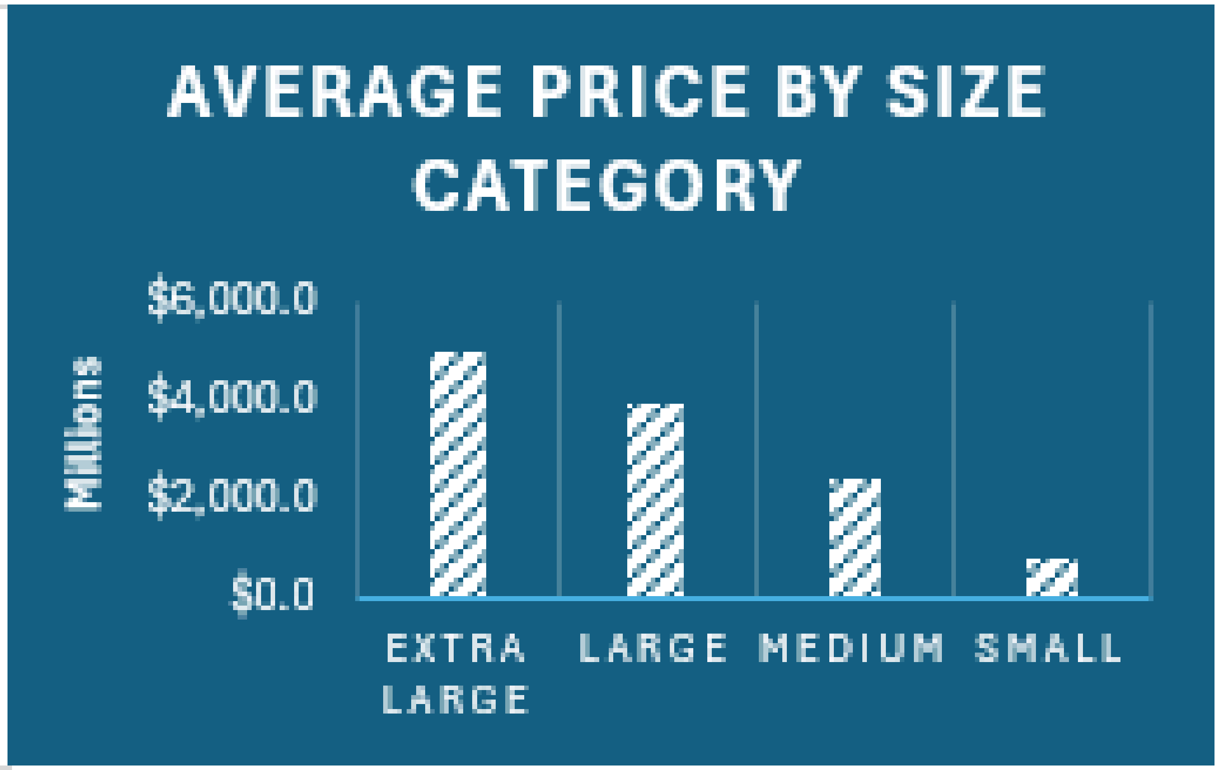
Scope: 10,004 residential property records.
Location: Paris, France.
Key Features: Square meters, room count, luxury add-ons (pool, yard), and historical data (year built, basement).

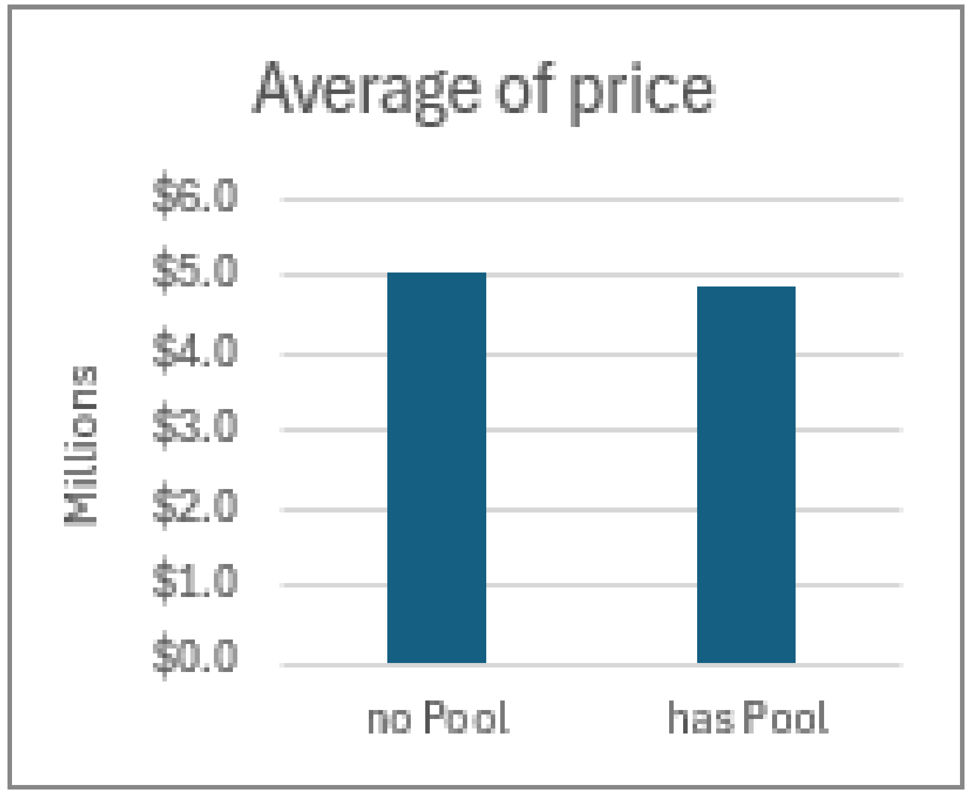
Paris Price House DashBoard

Average Price	Median Price	Average Size	Price per m2
$4,993,651.09	$5,016,631.00	49872.16 M2	$100.47



	price	squareM	#OfRooms	floors	rice_per_sM
price	1				
squareM	0.999999	1			
#OfRooms	-0.00233	-0.002333093	1		
floors	0.001808	0.001262439	-0.0075	1	
Price_per_sM	-0.28017	-0.280277805	-0.00382	0.045609	1





Size_Category: Extra ... | Large | Medium | Small

hasPool: has Pool | no Pool

hasStormProtector: 0 | 1

SPECIFY THE QUESTION

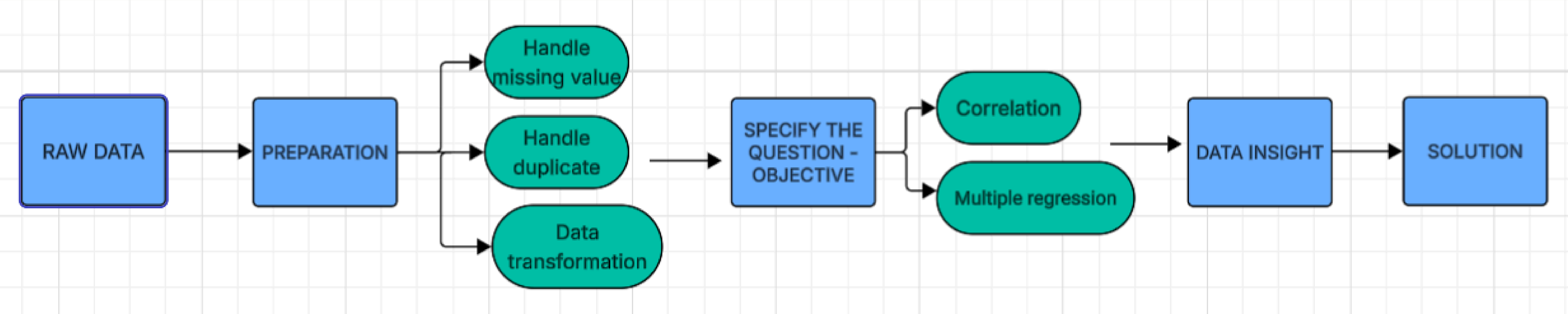
Question: What factors most significantly influence house prices in Paris?

Objective:
Identify pinpoint key features (square footage, luxury amenities, age) that significantly impact price.

Why this question is relevant?

- Maximizing Data Potential
- Market Precision
- Strategic Insight

FLOWCHART


RAW DATA
PREPARATION
Handle missing value
Handle duplicate
Data transformation
SPECIFY THE QUESTION - OBJECTIVE
Correlation
Multiple regression
DATA INSIGHT
SOLUTION

PREPARATION

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10004 entries, 0 to 10003
Data columns (total 13 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   squareMeters       10004 non-null  int64
 1   numberOfRooms      10003 non-null  float64
 2   hasYard            10004 non-null  int64
 3   hasPool            10004 non-null  int64
 4   floors             10004 non-null  int64
 5   numPrevOwners      10004 non-null  int64
 6   made               10004 non-null  int64
 7   isNewBuilt         10004 non-null  int64
 8   hasStormProtector  10004 non-null  int64
 9   basement           10004 non-null  int64
 10  garage             10004 non-null  int64
 11  hasStorageRoom     10004 non-null  int64
 12  price              10004 non-null  object
dtypes: float64(1), int64(11), object(1)
memory usage: 1016.2+ KB
```

HANDLE MISSING VALUE

The number of rooms is missing one value → Delete because it is small compared to all the data frames

PREPARATION

Handle duplicate

```
print(f'Shape of DataFrame before dropping duplicates: {df.shape}')
```

```
Shape of DataFrame before dropping duplicates: (10003, 13)
```

↓

```
df.drop_duplicates(inplace=True)
print(f'Shape of DataFrame after dropping duplicates: {df.shape}')
```

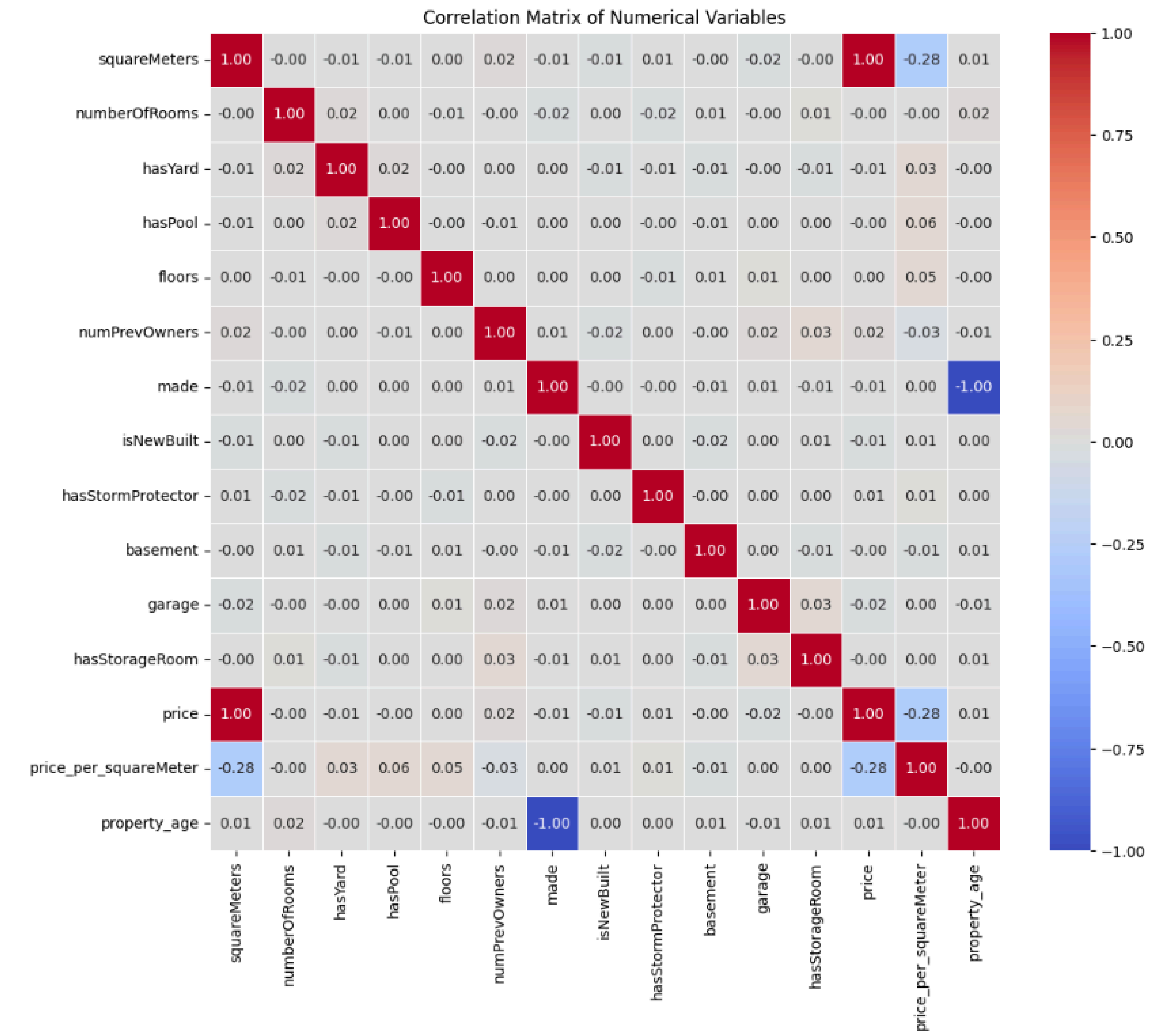
```
Shape of DataFrame after dropping duplicates: (10002, 13)
```

PREPARATION

Standardize the format of the Price column

```
<class 'pandas.core.frame.DataFrame'>
Index: 10003 entries, 1 to 10003
Data columns (total 13 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   squareMeters       10003 non-null  int64
 1   numberOfRooms      10003 non-null  float64
 2   hasYard            10003 non-null  int64
 3   hasPool            10003 non-null  int64
 4   floors             10003 non-null  int64
 5   numPrevOwners      10003 non-null  int64
 6   made               10003 non-null  int64
 7   isNewBuilt         10003 non-null  int64
 8   hasStormProtector  10003 non-null  int64
 9   basement           10003 non-null  int64
 10  garage             10003 non-null  int64
 11  hasStorageRoom     10003 non-null  int64
 12  price              10003 non-null  object
dtypes: float64(1), int64(11), object(1)
memory usage: 1.1+ MB
```

```
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   squareMeters       10003 non-null  int64
 1   numberOfRooms      10003 non-null  float64
 2   hasYard            10003 non-null  int64
 3   hasPool            10003 non-null  int64
 4   floors             10003 non-null  int64
 5   numPrevOwners      10003 non-null  int64
 6   made               10003 non-null  int64
 7   isNewBuilt         10003 non-null  int64
 8   hasStormProtector  10003 non-null  int64
 9   basement           10003 non-null  int64
 10  garage             10003 non-null  int64
 11  hasStorageRoom     10003 non-null  int64
 12  price              10003 non-null  float64
```



CORRELATION ANALYSIS

CORRELATION WITH PRICE

- Square Meters: nearly absolute correlation **(1.0)** with price - the only variable determining property value.
- Other factors: **low** correlation (<1)

REGRESSION ANALYSIS

Regression Statistics	
Multiple R	0.999999781
R Square	0.999999563
Adjusted R Square	0.999999562
Standard Error	1903.647476
Observations	10003

100% of price fluctuations

ANOVA

	df	SS	MS	F	Significance F
Regression	12	8.27971E+16	6.89976E+15	1.904E+09	0
Residual	9990	36202498405	3623873.714		
Total	10002	8.27971E+16			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	5013.675264	4104.27106	1.221575084	0.2218972	-3031.522931	13058.87346	-3031.522931	13058.87346
squareMeters	100.0000696	0.000661873	151086.5868	0	99.99877215	100.001367	99.99877215	100.001367
numberOfRooms	-0.765663893	6.64095917	-0.115294173	0.9082143	-13.78328187	12.25195408	-13.78328187	12.25195408
hasYard	3015.345275	38.08952409	79.16468758	0	2940.682134	3090.008417	2940.682134	3090.008417
hasPool	2982.157108	38.07602028	78.32113457	0	2907.520437	3056.793779	2907.520437	3056.793779
floors	54.54188278	0.659015975	82.76261099	0	53.25007869	55.83368687	53.25007869	55.83368687
numPrevOwners	0.200533776	6.67045112	0.030063001	0.9760174	-12.87489436	13.27596192	-12.87489436	13.27596192
made	-2.269982493	2.045569853	-1.109706662	0.2671522	-6.27971154	1.739746554	-6.27971154	1.739746554
isNewBuilt	158.274133	38.082799	4.156053051	3.265E-05	83.62417412	232.9240918	83.62417412	232.9240918
hasStormProtector	144.9527472	38.07774962	3.806757191	0.0001416	70.31268611	219.5928082	70.31268611	219.5928082
basement	-0.001415376	0.006618181	-0.213861852	0.8306592	-0.014388344	0.011557591	-0.014388344	0.011557591
garage	0.1141278	0.072710466	1.569619969	0.1165352	-0.028399364	0.256654964	-0.028399364	0.256654964
hasStorageRoom	16.2282376	38.11446351	0.425776362	0.67028	-58.48379007	90.94026528	-58.48379007	90.94026528

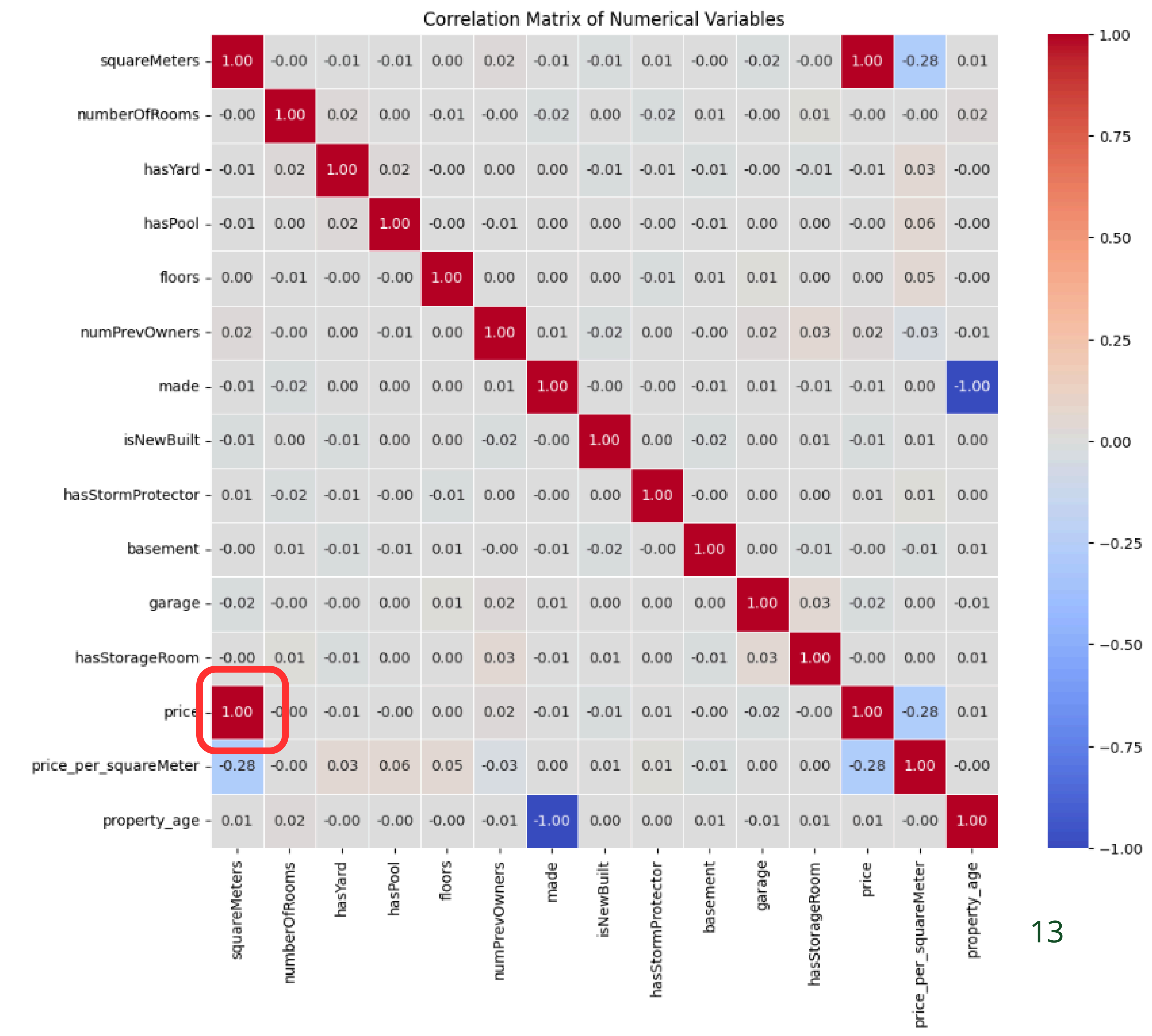
Product strategy: Focus on maximizing total usable space rather than dividing it into many smaller rooms.

Profit optimization: Investing in gardens and swimming pools is the quickest way to increase property value (besides expanding the area).

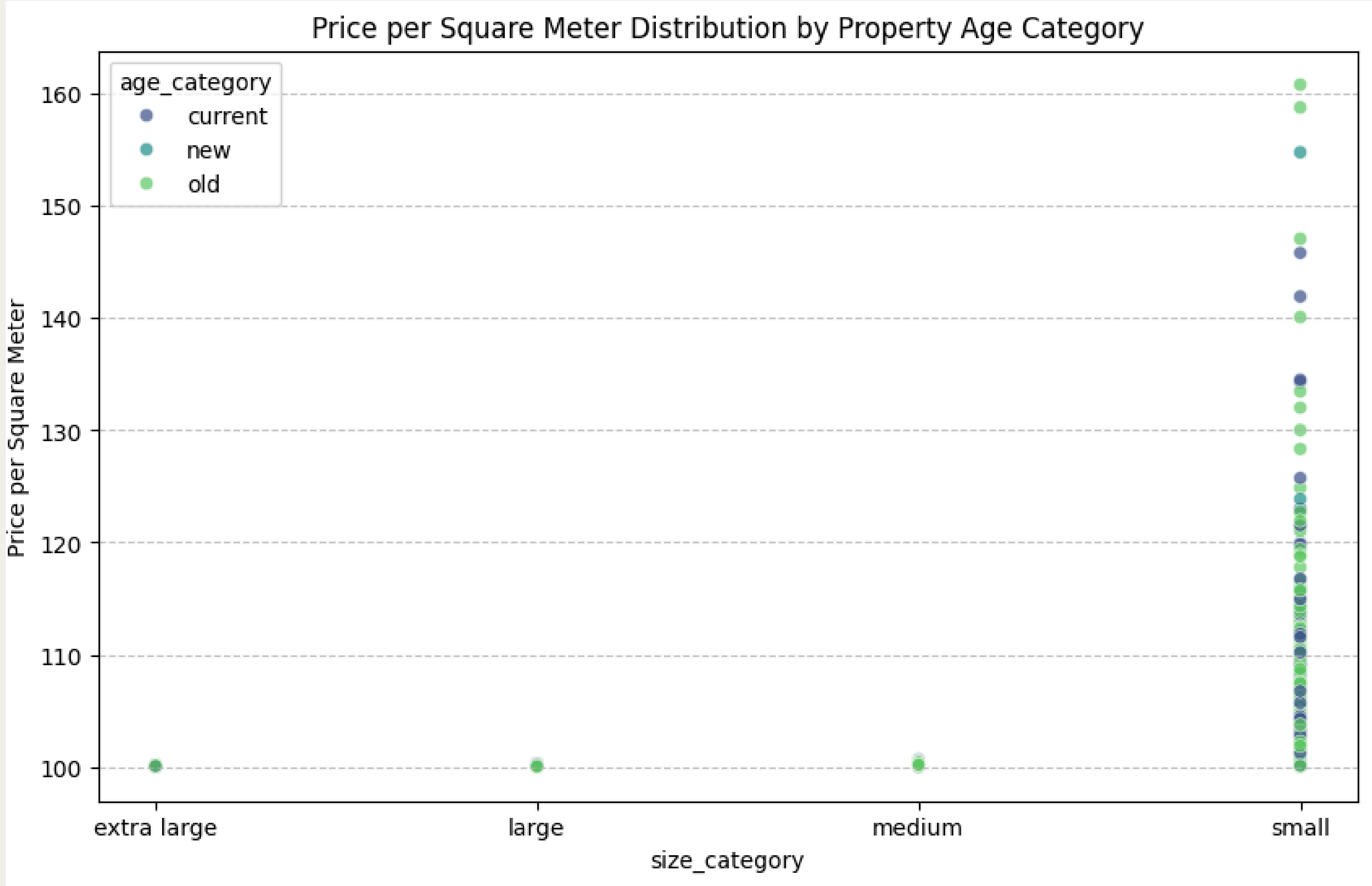
3.1 Insight 1: The Dominance of Square Meters

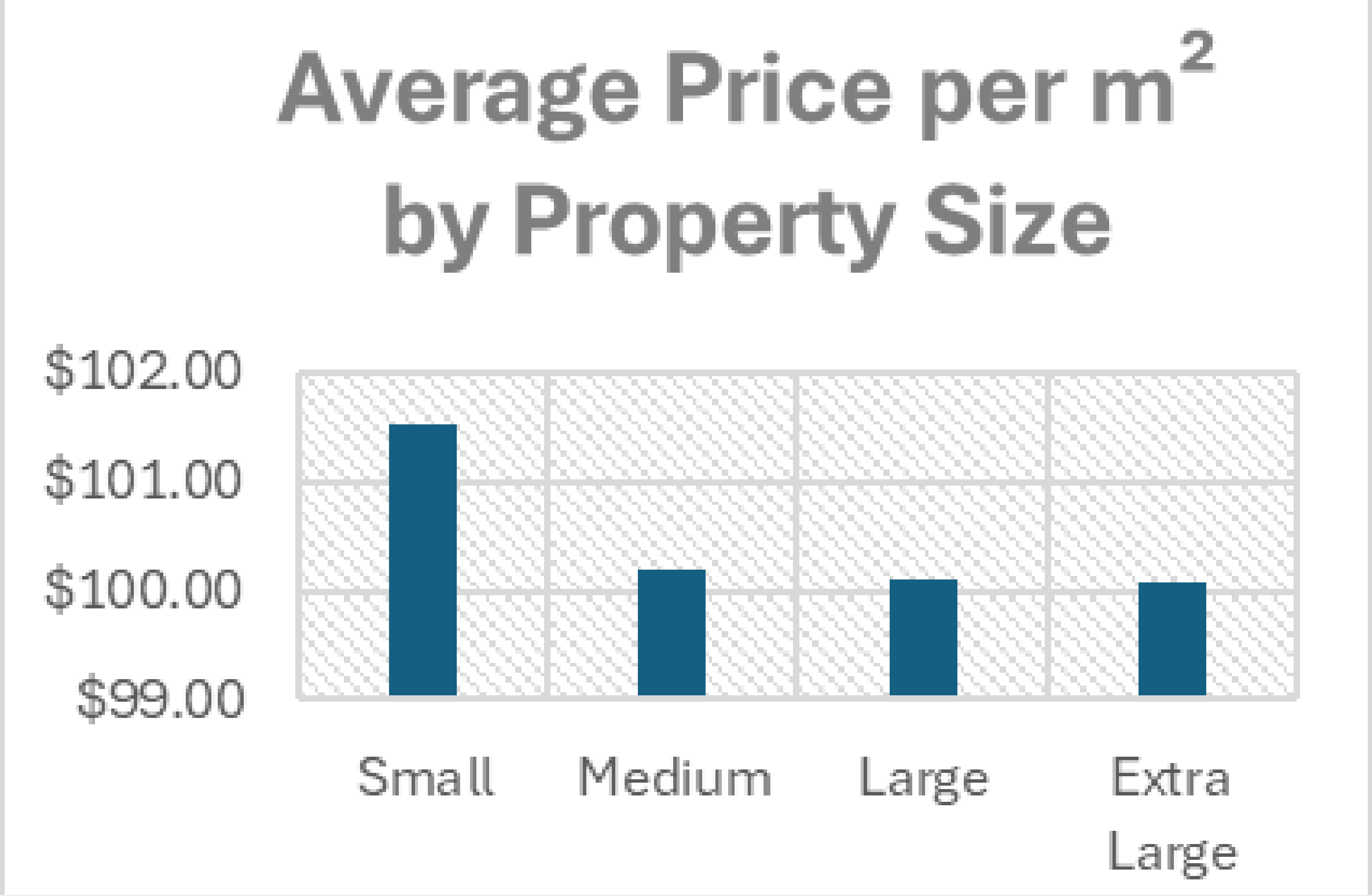
The square meter and price have a correlation coefficient of 1, indicating a strong positive correlation, meaning that house prices are entirely determined by the squaremeter factor.

→ Macro view



3.2 Insight 2: The "Micro-Pricing" Discovery





The 1.0 Illusion: Macro-correlation hides micro-pricing nuances.
Small Unit Premium: Unit prices (/m2) spike up to 10% for smaller areas.
Fixed Cost Impact: Property prices follow a non-linear "Base Cost" model.

→ **Micro view**

RECOMMENDATION

Adopt a "Buy in bulk" strategy.

Optimize space instead of indulging in luxury

Thank You!